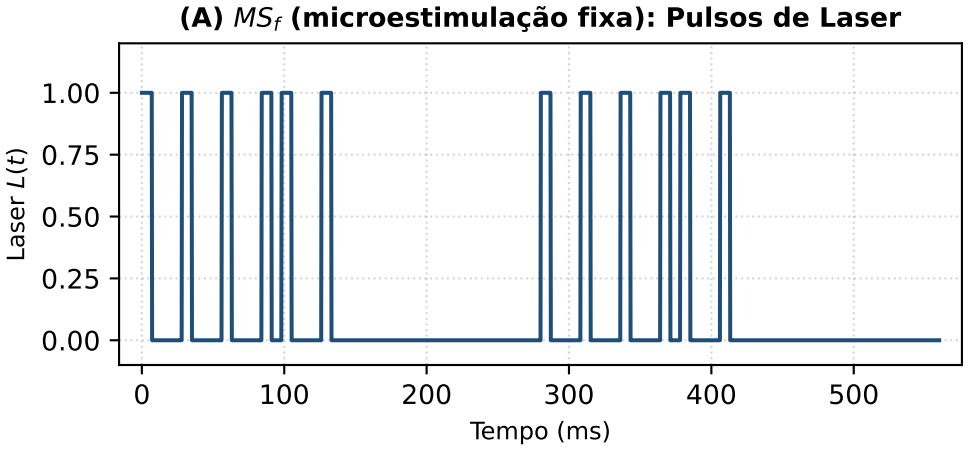
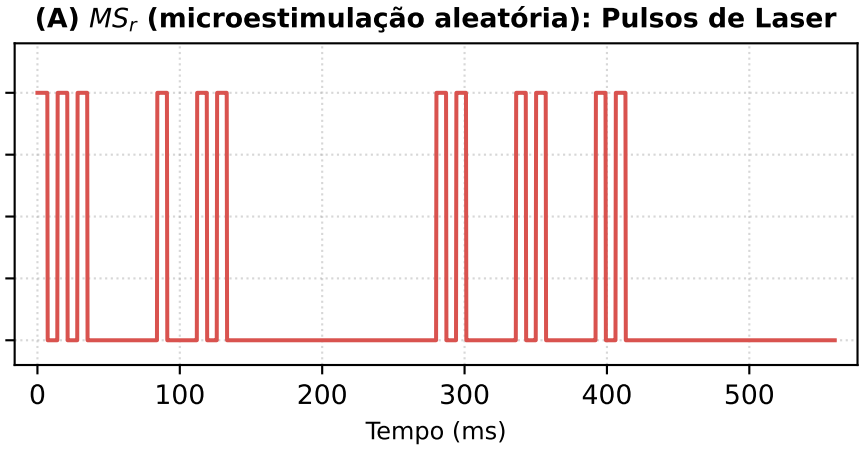
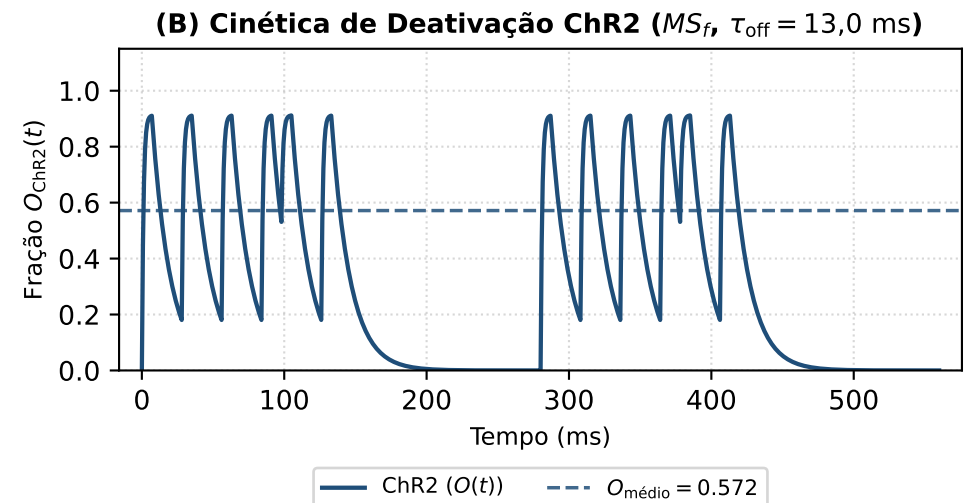
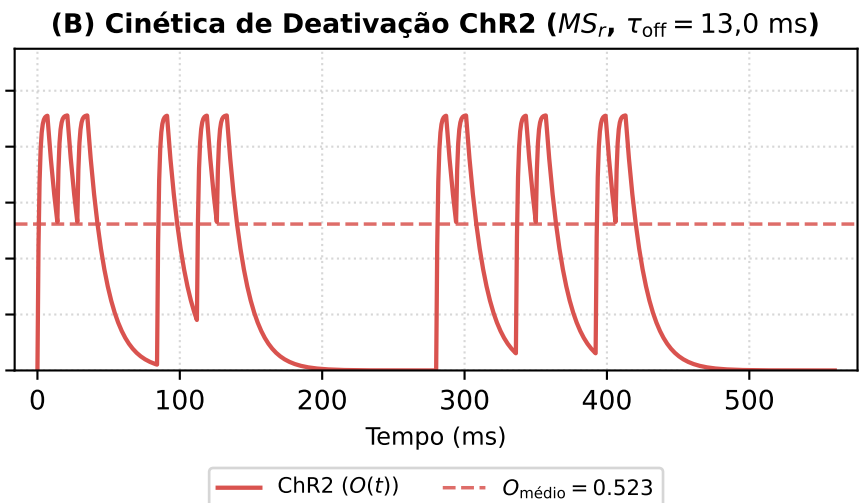
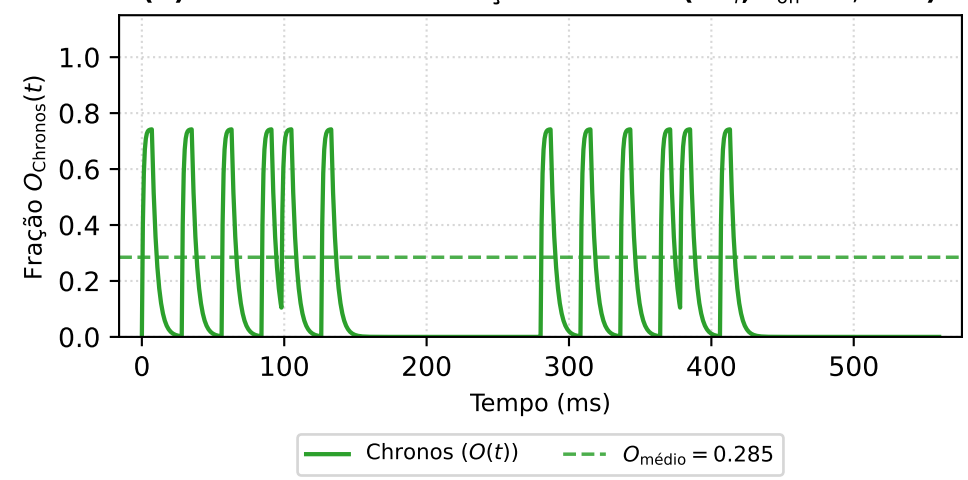
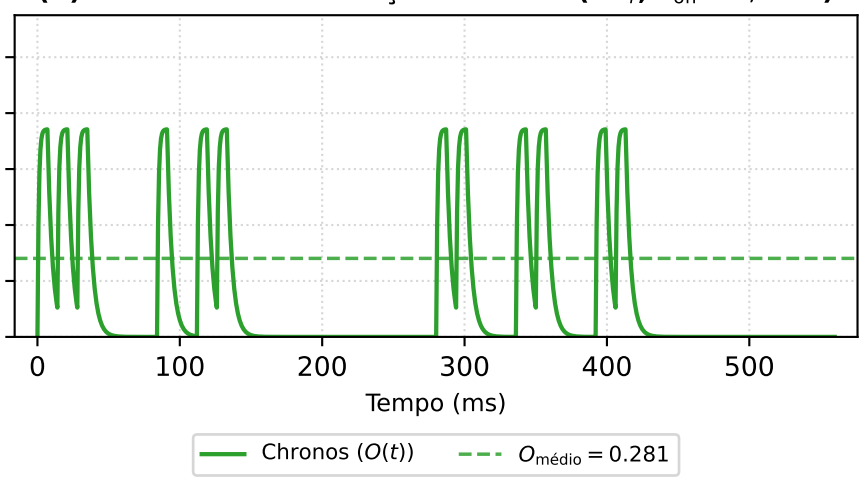

(A) MS_f (microestimulação fixa): Pulsos de Laser
Laser $L(t)$
1.00
0.75
0.50
0.25
0.00
0
100
200
300
400
500
Tempo (ms)

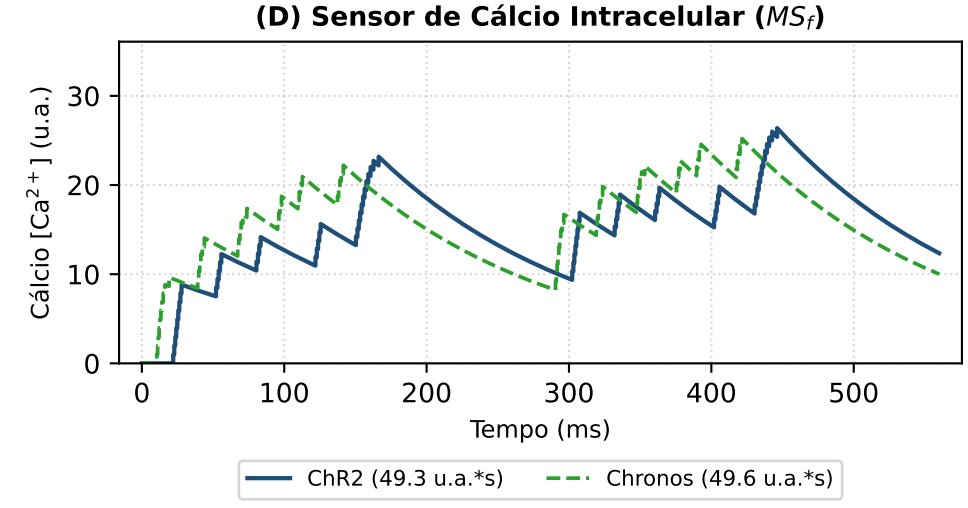

(A) MS_r (microestimulação aleatória): Pulsos de Laser
0
100
200
300
400
500
Tempo (ms)

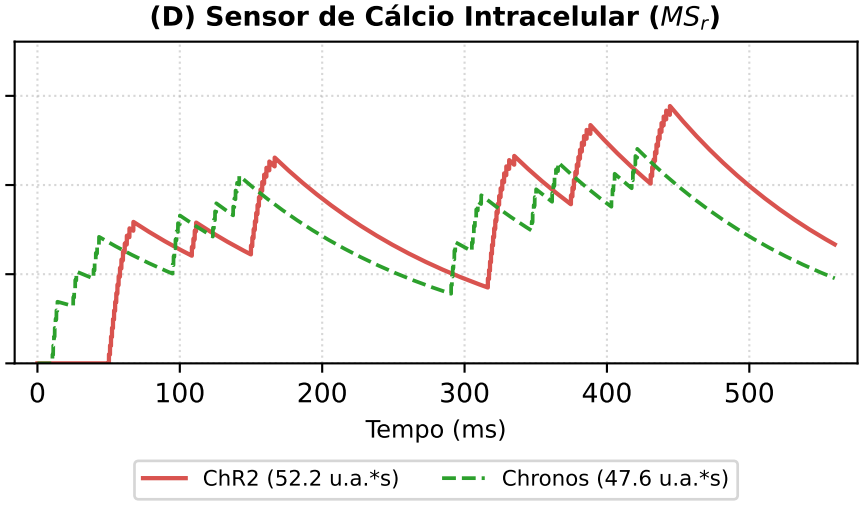

(B) Cinética de Deativação ChR2 (MS_f, $\tau_{off} = 13{,}0$ ms)
Fração $O_{ChR2}(t)$
1.0
0.8
0.6
0.4
0.2
0.0
0
100
200
300
400
500
Tempo (ms)
ChR2 ($O(t)$)
$O_{médio} = 0.572$


(B) Cinética de Deativação ChR2 (MS_r, $\tau_{off} = 13{,}0$ ms)
0
100
200
300
400
500
Tempo (ms)
ChR2 ($O(t)$)
$O_{médio} = 0.523$


(C) Cinética de Deativação Chronos (MS_f, $\tau_{off} = 3{,}6$ ms)
Fração $O_{Chronos}(t)$
1.0
0.8
0.6
0.4
0.2
0.0
0
100
200
300
400
500
Tempo (ms)
Chronos ($O(t)$)
$O_{médio} = 0.285$


(C) Cinética de Deativação Chronos (MS_r, $\tau_{off} = 3{,}6$ ms)
0
100
200
300
400
500
Tempo (ms)
Chronos ($O(t)$)
$O_{médio} = 0.281$


(D) Sensor de Cálcio Intracelular (MS_f)
Cálcio $[Ca^{2+}]$ (u.a.)
30
20
10
0
0
100
200
300
400
500
Tempo (ms)
ChR2 (49.3 u.a.*s)
Chronos (49.6 u.a.*s)


(D) Sensor de Cálcio Intracelular (MS_r)
0
100
200
300
400
500
Tempo (ms)
ChR2 (52.2 u.a.*s)
Chronos (47.6 u.a.*s)